One of the first DES systems that was investigated, contained ChCl and urea, in a ratio of 1:2 respectively.[105] There has also been a lot of work using carboxylic acids. Abbott *et al.* used phenylic carboxylic acids (phenylpropionic acid and phenylacetic acid) and showed that the eutectic composition with a quaternary ammonium salt was analogous to the urea ChCl case. This 2:1 ratio indicates that it requires two carboxylic acid molecules to complex with the chloride ion to form a eutectic mixture. However it was not possible to form DESs using monocarboxylic acids. By replacing the monocarboxylic acids with dicarboxylic acids with ChCl, there was a eutectic point reached and this eutectic was formed at a ratio of 1:1. This reinforces the theory that each anion is complexed by two hydrogen bonds.[109] It has also been shown that tricarboxylic acids can be used as the hydrogen bond donor. The eutectic point is reached when the composition is between 30-35 molar percentage composition of the acid. Because there are so many hydrogen bond donating groups in these systems, they exhibit the rheology of gels. This is thought that there is extensive bridging of the acids between neighbouring chloride ions. There is also considerable evidence for the depression in the freezing points of the system being in part governed by the molecular weight of the acid, with smaller molecular weight acids exhibiting greater depression in the freezing points. In general, the freezing point of any mixture containing a quaternary ammonium salt and a hydrogen bond donor is dependent on many things but the two most important factors are firstly, how large the individual lattice energies of both the hydrogen bond donor and the quaternary ammonium salt is and secondly the degree to which the anion-hydrogen bond donor interact. If the anion-hydrogen bond donor interaction is strong, the entropy of the system increases, resulting in a more disordered system, therefore offering a lower freezing point. To a certain degree, the depression in the freezing point can be a measure of the entropy change.

These types of DES have been used very successfully for many purposes such as electrochemical applications. Liquids such as Maline 100 (ChCl : malonic acid) and Ethaline 200 (ChCl : 2 x 1,2-ethanediol) have generated a lot of industrial interest in their use for both polishing and plating of metals.

1.6.4 Anhydrous Metal Halide Substituted Hydrogen Bond Donor Mixtures

This is a relatively new area of research in the DESs area.[113] The idea of these systems is to use the metal halide, which in Section 1.6.3 was shown to be the complexing species that disrupts the interaction of the anion/cation interaction of the quaternary ammonium salt, the hydrogen bond donor reverses the role of it. Essentially the metal halide is the replacement for the quaternary ammonium salt. The hydrogen bond donor complexes to one or more of the halides of the metal and withdraws it from the metal centre. The majority of work in this area has been using $ZnCl_2$ as the metal halide and hydrogen bond donors that have been used are acetamide, 1,6-hexanediol, urea and 1,2-ethanediol.[113] Again, as described in Section 1.6.3., the eutectic point is not always as obvious as counting two hydrogen bond donations per halide ion. This is seen with the case for $ZnCl_2$ with both acetamide and 1,2-ethanediol (1:4), but with 1,6-hexandeiol, the eutectic forms at 1:3. This is most probably down to the strength and size of the hydrogen bond donating species, and also the form of the metal halide species.

1.7 The Use of Choline Chloride (hydroxyethyltrimethylammonium chloride) As The Common Quaternary Ammonium Salt

In the area of DESs, the most common quaternary ammonium salt to be used is choline chloride. Initially, this was because of the physical properties it offered when used as part of a DESs. When mixed with most hydrogen bond donors or metal halides, the physical properties were almost always greatly improved over the use of most other quaternary ammonium salts. The depression in freezing point tended to be one of the largest, if not the largest. An example of this is the ChCl: urea mixture. By themselves they have freezing points of 303 °C and 135 °C respectively. It also showed lower viscosities than many other systems containing different quaternary ammonium salts and conductivites were often higher when ChCl was used. The main reason behind ChCl being such a useful quaternary ammonium salt is to do with the fact it is an asymmetric quaternary ammonium salt with a polar functional group, but is also small. The asymmetric nature of this molecule reduces the freezing point of the ionic-

molecular liquid, as does the polar functional group. By combining these two compounds in a ratio of 1:2 (ChCl: urea), the product formed has a freezing point of 12 °C.

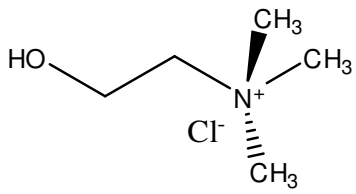


***Figure 1.9**: Structure of Choline Chloride (hydroxyethyltrimethylammonium chloride)*

Perhaps in terms of public perception of this being used as a part of a potential replacement for convential solvents, the fact that choline chloride is essentially non-toxic is a large plus point for its use. Choline chloride is also known as pro-vitamin B4 and has a recommended daily allowance of ~500 mg. It has been used for many years primarily as a chicken feed additive so is very cheap to buy. With the current economic and industrial explosion occurring in China, it has become possible to buy it from Chinese chemical companies for almost £1 per Kg.

1.8 References

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CHAPTER 2

Experimental

2.1 Materials

2.2 Purification of Choline Chloride

2.3 Physical Properties Experiments

2.3.1 Synthesis of ChCl: alcohol systems

2.3.1.1 Synthesis of 33%:67% ChCl: 1,2-Ethanediol system

2.3.1.2 Synthesis of 25%:75% ChCl: 1,4-Butanediol system

2.3.1.3 Synthesis of 25%:75% ChCl: 1,6-Hexanediol system

2.3.1.4 Synthesis of 20%:80% ChCl: 1,3-Butanediol system

2.3.1.5 Synthesis of 20%:80% ChCl: 1,2-Butanediol system

2.3.1.6 Synthesis of 20%:80% ChCl: 2,3-Butanediol system

2.3.1.7 Synthesis of 33%:67% ChCl: Glycerol system

2.3.2 Viscosity Measurements

2.3.3 Conductivity Measurements

2.3.4 Surface Tension Measurements

2.3.5 Density Measurements

2.4 Solvatochromism Experiments

2.5 Esterification Experiments

2.5.1 Synthesis of 1:3 ChCl: Glycerol System for Estericaton Reaction

2.5.2 Gel Permeation Chromotography Analysis of Estericaton Reactions

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2. Experimental

2.1 Materials

Table 2.1: List of compounds used during the project

Compound	Purity	Molecular Weight	Melting Point
1,2-ethanediol	99%	62.07	-13 ºC
1,2-butanediol	99%	90.12	-50 ºC
1,3-butanediol	99%	90.12	0 ºC
1,4-butanediol	99%	90.12	16 ºC
2,3-butanediol	98%	90.12	25 ºC
choline chloride	99%	139.63	303-305 ºC
***meso*-erythritol**	99%	122.12	118-120 ºC
D-(-)fructose	99%	180.16	119-122 ºC
D-glucose	99%	180.16	150-152 ºC
glycerol	99%	92.09	18 ºC
1,6-hexanediol	98%	118.18	43-45 ºC
lauric acid	99%	200.31	44-46 ºC
N,N,-dimethyl-4-nitroaniline	99%	166.18	163-165ºC
4-nitroaniline	99%	138.13	146ºC
phenol	99%	94.11	40-42 ºC
Reichardt's Dye	90% +	551.68	271-275 ºC
trifluoroacetic Acid	98%	114.02	-15 ºC
xylitol	99%	152.15	94-97 ºC

2.2 Purification of Choline Chloride

Choline chloride (ChCl) is recrystallised to obtain the highest purity possible to be used in all experiments. Approximately for every 100 grams of ChCl, 100 mL of ethanol is used to recrystallise the ChCl. The solution is heated under reflux for approximately 2 hours, by which time all the choline chloride has dissolved, hot filtered,

and the filtrate allowed to cool. Upon cooling crystalline choline chloride reforms and the impurities stay dissolved in the ethanol. The crystalline product is vacuum filtered using a large sintered funnel and dried and kept under vacuum. Pure white crystals are obtained. Purity was determined by melting/decomposition point which was in the range of 302-305 °C, which corresponds to the know decomposition point of ChCl.

2.3 Physical Properties Experiments

2.3.1 Synthesis of ChCl: alcohol systems

The procedure for the formation of selected systems used in Chapter 3, are as follows:

2.3.1.1 Synthesis of 33%:67% ChCl: 1,2-Ethanediol system

To 20 cm^3, (22.26 g, 0.359 mol) of 1,2-ethanediol, 24.664 g (0.177 mol) of ChCl was added and heated at ~ 60 ºC, until fully dissolved. When the ChCl was fully dissolved the mixture was then cooled to the appropriate temperature before any measurements were taken.

2.3.1.2 Synthesis of 25%:75% ChCl: 1,4-Butanediol system

To 20 cm^3, (20.34 g, 0.226 mol) of 1,4-butanediol, 10.505 g (0.075 mol) of ChCl was added, it was stirred and heated at ~ 60 ºC, until fully dissolved. When the ChCl was fully dissolved the mixture was then cooled to the appropriate temperature before any measurements were taken.

2.3.1.3 Synthesis of 25%:75% ChCl: 1,6-Hexanediol system

To 20 g, (0.254 mol) of 1,6-hexanediol, 7.877 g (0.056 mol) of ChCl was added, it was stirred and heated at ~ 80 ºC, until fully dissolved. When the ChCl was fully dissolved the mixture was then cooled to the appropriate temperature before any measurements were taken.

2.3.1.4 Synthesis of 20%:80% ChCl: 1,3-Butanediol system

To 20 cm^3, (20.1 g, 0.223 mol) of 1,3-butanediol, 7.786 g (0.056 mol) of ChCl was added, it was stirred and heated at ~ 60 ºC, until fully dissolved. When the mixture was fully homogenous, the mixture was then cooled to the appropriate temperature before any measurements were taken.

2.3.1.5 Synthesis of 20%:80% ChCl: 1,2-Butanediol system

To 20 cm^3, (20.12 g, 0.223 mol) of 1,2-butanediol, 7.793 g (0.056 mol) of ChCl was added, it was stirred and heated at ~ 60 ºC, until fully dissolved. When the mixture was fully homogenous, the mixture was then cooled to the appropriate temperature before any measurements were taken.

2.3.1.6 Synthesis of 20%:80% ChCl: 2,3-Butanediol system

To 20 cm^3, (19.9 g, 0.221 mol) of 2,3-butanediol, 7.708 g (0.052 mol) of ChCl was added, it was stirred and heated at ~ 60 ºC, until fully dissolved. When the mixture was fully homogenous, the mixture was then cooled to the appropriate temperature before any measurements were taken.

2.3.1.7 Synthesis of 33%:67% ChCl: Glycerol system

To 20 cm^3, (25.12 g, 0.273 mol) of glycerol, 19.041 g (0.136 mol) of ChCl was added, it was stirred and heated at ~ 80 ºC, until fully dissolved. When the mixture was fully homogenous, the mixture was then cooled to the appropriate temperature before any measurements were taken.

2.3.2 Viscosity Measurements

The viscosity of various eutectic mixtures was measured using a Brookfield DV-E Viscometer fitted with a thermostated jacket. The viscometer jacket was connected to a

thermostated water bath. The water bath was always approximately 0.5 ^{o}C warmer than the sample in the viscometer so the thermostat of the water bath was always set at 0.5 ^{o}C higher than the desired temperature. The viscosity measurements were obtained using a spindle attachment, which measures the ease at which the spindle can move at a set amount of revolutions per minute. Readings were taken after 25-30 minutes for each temperature to ensure good equilibration of temperature between the water bath and the sample.

2.3.3 Conductivity Measurements

The conductivity of various eutectic mixtures was measured using a Jenway 4071 Conductivity Meter conductivity probe. Measurements were carried out at exactly the same temperatures as the viscosity experiments were for the same eutectic mixture. To ensure good correlation between the results, the sample used for conductivity was made from the same batch as that used in the viscosity experiments. The readings were taken after 25-30 minutes for each temperature selected to ensure good equilibration of temperature between the water bath and the sample.

2.3.4 Surface Tension Measurements

The surface tension of various eutectic mixtures was measured using a Krüss Tensiometer K9 with a platinum plate. The plate is dipped into the liquid so it is just touching the surface and the resulting reading is recorded and then the plate is removed from the liquid.

2.3.5 Density Measurements

The densities of the systems are measured on the Krüss Tensiometer K9. The density probe is submerged into the liquid for 5 minutes so the apparatus can reach equilibrium, and then the reading on the tensiometer is taken.

2.4 Solvatochromism Experiments

All the systems investigated in Chapter 4 were made in an analogous way to those in Chapter 3. A Shimadzu Model UV-1601 Spectrophotometer was used to measure the solvatochromic shift of the different indicator dyes in the visible absorbance spectrum. The path length was 1 cm. The program UV PROBE was used to obtain the maximum absorbance of the UV-Vis spectra of the solvatochromic dyes in each eutectic mixture. The spectrum of the solvatochromic probe in the systems were measured at a resolution of 0.05 nm per data point. The estimated uncertainty in the wavelength maximum is less than 0.5 nm. The concentration of samples were kept in the range of 10^{-5} – 10^{-6} mol dm^{-3} in order to reduce any possible solute-solute interactions being observed.

2.5 Esterification Experiments

In Chapter 5, it is mentioned that various molar amounts of lauric acid are used. The following data are the exact amounts of lauric acid added to the systems for each data set.

Table 2.1: *Weights of lauric acid used*

Molar amount of lauric acid	**0.1 M**	**0.2 M**	**0.3 M**	**0.4 M**	**0.6 M**
Weight of lauric acid	20.032 g	40.064 g	60.096 g	80.128 g	120.192 g

2.5.1 Synthesis of 1:3 ChCl: Glycerol System for Esterication Reaction

To 60 cm^3, (75.36 g, 0.82 mol) of glycerol, 38.084 g (0.27 mol) of ChCl was added and heated at ~ 80 ºC, until fully dissolved. When dissolved the mixture was heated to 150 ºC. At this point the lauric acid and 6 cm^3 (9.21g, 0.08 mol) trifluoroacetic acid catalyst was added and then sealed in the reaction vessel. The mixture was homogenised using a mechanical stirrer at 800 rpm. All other systems were made up in analogous method to this.

2.5.2 Gel Permeation Chromotography Analysis of Estericaton Reactions

The progression of the esterification reactions was measured by gel permeation chromatography (GPC). The GPC used was a Polymer Laboratories PL-GPC 220 with two PLgel 5µm 50Å columns. The analysis was carried out at 40 °C at a flow rate of 1 cm/min. The solvent used was still dried THF. Each sample vial contained 0.02 g of product and was dissolved in 1.25 mL of dry THF. Under this setup the elution times of the reagents and products are shown in Figure 2.2.

Figure 2.2: *Peak elution time of reagents and products of esterification reactions using GPC*

Reagent/Product	Lauric Acid	Mono Ester	Di Ester	Tri Ester
Elution Time /min	14.7	14.2	13.3	12.9

CHAPTER 3

Physical Properties of Alcohol Based Deep Eutectic Solvents

3.1 Variation of the Chain Length of the Hydrogen Bond Donor

3.1.1 1,2-Ethanediol

3.1.2 1,4-Butanediol and 1,6-Hexanediol

3.2 Effects Hydrogen Bond Donor Group Positioning

3.3 Variation of the Number of Hydrogen Bond Donating Groups

3.3.1. Physical Properties of Glycerol with ChCl

3.4 At Which Composition do DESs Become Ionic Liquids?

3.5 Poly-Alcohol/Sugar Based Deep Eutectic Solvents

3.6 Mono Functional Alcohol Deep Eutectic Solvents

3.7 Conclusions

3.8 References

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3. Introduction

"Deep Eutectic Solvents" can be described by the following general formula:

$$\underbrace{(R_1R_2R_3R_4N^+)}_{(A)}\ \underbrace{(X^-)}_{(B)}\ \underbrace{(R_5)}_{(C)}\underbrace{(YH)}_{(D)} \qquad (3.1)$$

This general formula encompasses in excess of 10^6 possibilities and clearly not all variants can be studied. The aim of this part of the project is to characterise the effect of elements C and D in equation (3.1). For this study, only choline chloride (ChCl) has been used as the quaternary ammonium salt. Other cations and anions have previously been studied and some basic models of phase behaviour have been developed. ChCl has been chosen as it is the most commonly used precursor for eutectic based ionic liquids and already has a number of industrial applications. It offers optimal environmental and cost benefits and can therefore be applied to large-scale processes.

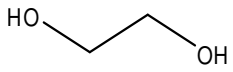
Eutectic mixtures of ChCl with ethylene glycol have already been applied to the electropolishing of stainless steel,[1,2] the plating of metals and alloys,[3] the electroless deposition of silver on copper[4,5] and the recovery of metals from complex metal oxide matrices.[6] and understanding of the phase behaviour, viscosity, conductivity and density would therefore be useful for the development of new and improved DESs.

3.1 Variation of the Chain Length of the Hydrogen Bond Donor

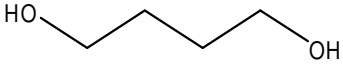
In this section R_5 from the general formula (3.1) is varied. So far there has been a lot of work in the variation of the hydrogen bond donor (C) and (D). Previously various dicarboxylic acids and diamides have been studied but very little work has been carried out using diols. This section will investigate various glycols, which will alter the variable (C) but keep variable (D) constant. The length of the chain of the glycol was altered to determine the effect this has on the physical properties of the system.

The viscosity, conductivity, surface tension and density were measured as both a function of temperature and a function of molar composition. The glycols used were 1,2-ethanediol (5), 1,4-butanediol (6) and 1,6-hexanediol (7). The estimated error in these measurements is believed be approximately +/- 1% due intrinsic errors from the

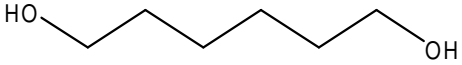
instrumentation, which is approximately the same size as the data markers, so have not been included.



(5)



(6)



(7)

3.1.1 1,2-Ethanediol

Mixtures of ChCl with 1,2-ethanediol (5) were made ranging from 5% to 33% ChCl. The viscosity of theses systems show a decrease exponentially as the temperature is increased due to increased kinetic energy. By adding more ChCl the viscosity of the systems increase as would be expected. This can be seen in Appendix Figure 7.1. The viscosity of these liquids is significantly less than other eutectic based ionic liquids as can be seen from Figure 3.1.

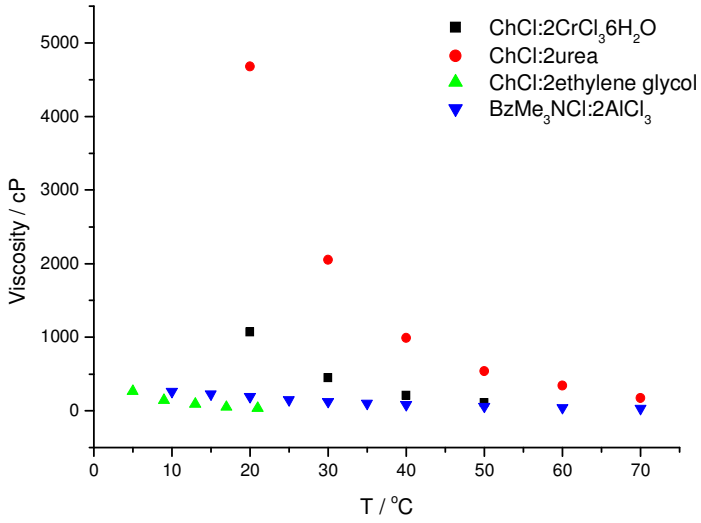


***Figure 3.1**: Viscosity of various eutectic based ionic liquids as a function of temperature redrawn from the literature*[7-9]

The viscosity of an ionic liquid has been fitted to two types of model; one based on an Arrhenius model and one based on the Vogel–Tamman–Fulcher (VTF) model. In this work the former model will be applied as suitable correlation is observed and it allows correlation to previously reported data. The energy of activation of viscous flow (E_η) of each of the systems was obtained by plotting a graph of ln viscosity versus inverse temperature according to equation 3.2:

$$\ln \eta = \ln \eta_0 + \frac{E_\eta}{RT} \quad \textbf{(3.2)}$$

where η_0 is a constant.

The data are shown in Figure 3.2 and the values of both E_η and $\ln \eta_0$ are reported in **Table 3.1** along with the values obtained for the other glycols studied.

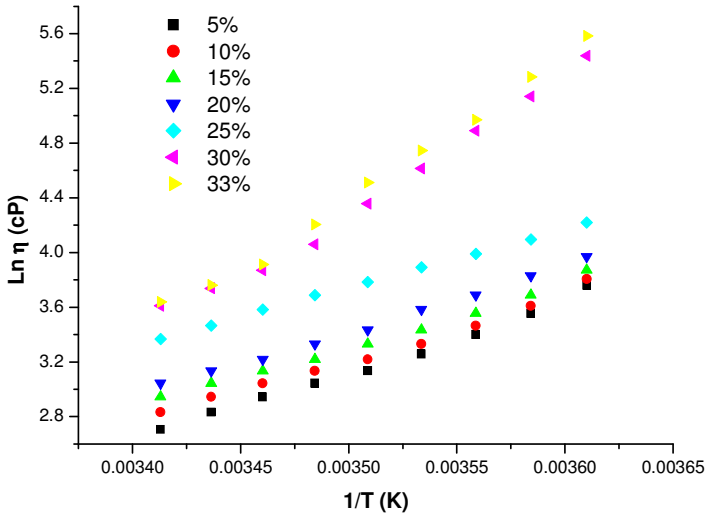


***Figure 3.2**: Plot of inverse temperature versus log viscosity as a function of ChCl composition in 1,2-ethanediol*

The conductivity, κ, for the systems shown in Figure 3.2 were also obtained and these are shown in Appendix Figure 7.2. In an analogous manner to the viscosity data the conductivity data can be fitted to an Arrhenius based equation of the form

$$\ln \kappa = \ln \kappa_0 - \frac{E_\Lambda}{RT} \quad \textbf{(3.3)}$$

where E_Λ is the energy of activation of conductivity and κ_0 is a constant, by plotting ln conductivity versus inverse temperature, as can be seen in Figure 3.3.

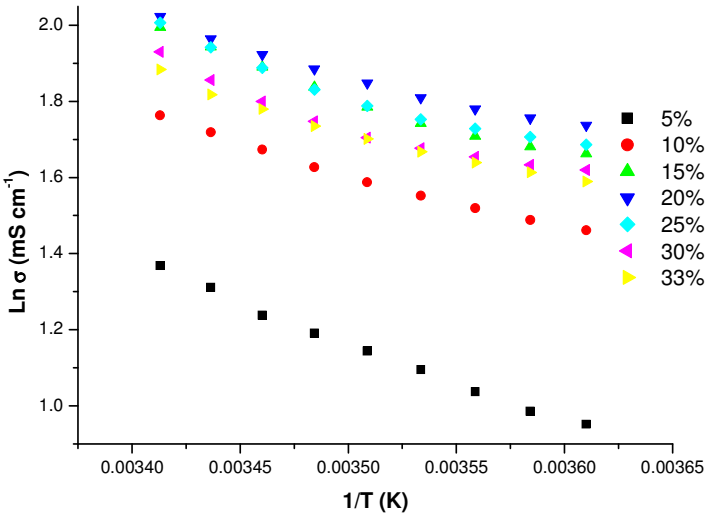


***Figure 3.3**: Plot of inverse temperature versus log conductivity as a function of ChCl composition in 1,2-ethanediol*

The data obtained for conductivity offers some interesting information on these systems. In the case of viscosity as a function of composition, there was an increase in the viscosity as more ChCl was added. The observed trend in conductivity shows an increases from 5% to 20% as would be expected since more charge carrying species is being added to the system, however from 20% to 33% there is a decrease in the conductivity. This can be seen more clearly in Figure 3.4 by just plotting the data at 293 K.

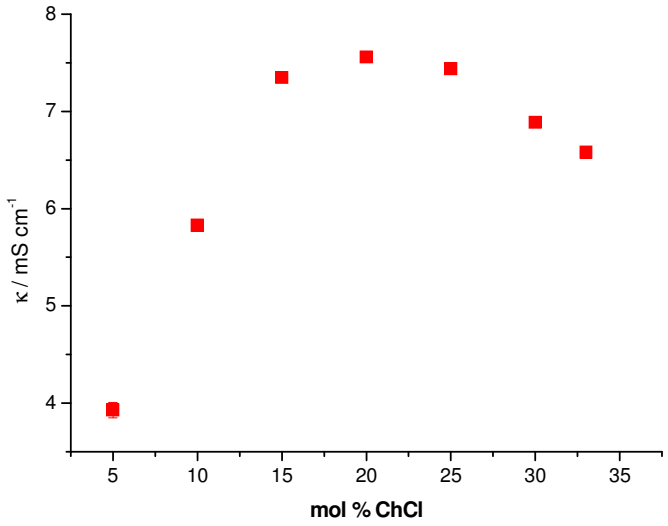


***Figure 3.4**: Plot of molar percentage composition of ChCl in 1,2-ethanediol versus conductivity at 293K*

This is believed to be due to the effect of viscosity being greater than the effect of conductivity on the system after more than 20 % ChCl is added. Despite the fact there are more charge carrying species (ChCl) in the system, the viscosity of the system impedes the flow of ions through the system. Probably the most important observation about the data in Figure 3.3 however is the magnitude of the conductivity which is significantly greater than the other ChCl based eutectics reported previously.[7,8] The data are also comparable with some of the highest conductivities reported for room temperature ionic liquids with discrete anions.[10-13] This suggests that the motion of ions in these liquids may be different to other choline based eutectics.

The conductivity of concentrated quaternary ammonium electrolytes in non-aqueous solutions has previously been modelled using the Fuoss Kraus theory.[14] The molar conductivity, Λ is given by

$$\Lambda = \frac{\Lambda^0 \sqrt{K_p}}{\sqrt{c}} + \frac{\Lambda_t^0 \sqrt{K_p c}}{K_t} \qquad \textbf{(3.4)}$$

Where Λ^0 and Λ^0_t are the limiting molar conductivity of the single and triple ions respectively, c is the concentration of the electrolyte and K_p and K_t are the ion pair and triple ion dissociation constants respectively. This has subsequently been modified to account for the activity coefficient of the electrolyte, but this theory is only valid at relatively low concentrations (< 0.1 mol dm^{-3}). It was proposed that in these systems

ion-ion interactions were strong and ion pairs and triple ions dominated charge transport.

The mobility of ions in ionic liquids is significantly different from that in non-aqueous solvents and there have been very few attempts to model the conductivity of ionic liquids. The Walden rule ($\Lambda\eta = constant$) has been applied to ionic liquids, where Λ is the molar conductivity and η is the viscosity.[14] Deviations from the Walden rule seen in proton transfer ionic liquids have previously been used to explain ionic association.[15,16] This rule is not fully applicable to ionic liquids as it is only valid for ions at infinite dilution where ion-ion interactions can be ignored, which is clearly not the case in ionic liquids.

Hole theory has been used to analyse the viscosity of ionic liquids.[17] It was initially developed by Fürth[18] in an attempt to explain the properties of the liquid state and was later expanded by Bockris *et al.* for use with molten salts.[19,20] This model was also found to have serious limitations in molten salts because the ionic association significantly affects the ionic activity. Stillinger has provided a discussion of the various models for pure fused salts.[21]

Abbott *et al*. recently showed that E_η is related to the size of the ions and the size of the voids present in the liquid for ionic liquids with discrete anions and DESs.[7,17,22] It was shown that the viscosity of ambient temperature molten salts is several orders of magnitude higher than high temperature molten salts. This was due partially to the difference in size of the ions, but also due to the increased void volume in the liquid.

Abbott later showed that hole theory can be applied to both ionic and molecular fluids to account for viscosity.[17,22] It has been shown that the viscosity of a fluid, η, can be modelled by assuming it behaves like an ideal gas. Its motion however is restricted by the availability of sites of the ions/molecules to move into. Hence it was shown that

$$\eta = \frac{m\bar{c} / 2.12\sigma}{P(r > R)} \qquad \textbf{(3.5)}$$

Where m is the molecular mass (for ionic fluids this was taken as the geometric mean), $\bar{c}$ is the average speed of the molecule ($=(8kT/\pi m)^{1/2}$) and σ is the collision diameter of the molecule ($4\pi R^2$). The probability of finding a hole of radius, r, greater than the radius of the solvent molecule, R, in a given liquid, $(P(r > R))$, is given by integration of the following expression[24];

$$Pdr = \frac{16}{15\sqrt{\pi}} a^{7/2} r^6 e^{-ar^2} dr \qquad \textbf{(3.6)}$$

where $a = 4\pi\gamma / kT$ and γ is the surface tension. There is good correlation obtained between the calculated and measured viscosities, which show that the viscosity of fluids as being limited by the availability of holes large enough for the mobile species to be able to move into. It should be noted that inspection of equations (3.5) and (3.6) shows that a decrease in the viscosity can be obtained by decreasing the surface tension of the liquid i.e. increasing the free volume, or by decreasing the ionic radius.

There is comparatively little surface tension data reported on ionic liquids but work by Branco *et al.*[24] and Huddleston *et al.*[25] showed that ionic liquids with fluorinated anions have low surface tensions and low viscosities. This explains why the ionic liquids with highly fluorinated anions exhibit the lowest viscosities. The effect of the cation on viscosity however is less clear. In the case of the imidazolium cation, an increase in the length of the R group should result in a decrease in the viscosity as the degree to which the ion-ion interactions occur will be reduced and hence the surface tension will decrease. On the other hand however, the increase in size of the alkyl group will reduce its mobility due to a lack of suitably sized voids for the larger cations to move into. This explains the data presented by Tokuda *et al.* who showed a minimum in viscosity for ethyl methyl imidazolium salts.[26]

Since it has been shown that viscosity is limited by the availability of holes large enough for the mobile species to move into, then charge mobility can be considered as being equal to the migration of holes in the opposite direction to that of the ions. The fraction of suitably sized holes in ambient temperature ionic liquids is very low (*c.a.* 10^{-6}), therefore the holes can be assumed to be at infinite dilution. The migration of the mobile species should therefore be described by a combination of the Stokes-Einstein and Nernst Einstein equations

$$\lambda_+ = z^2 F e / 6 \pi \eta R_+ \qquad \textbf{(3.7)}$$

where z is the charge on the ion, F is the Faraday constant and e is the electronic charge. With the Stokes-Einstein equation being valid for both the cation and anion, the conductivity of the salt can be determined since

$$\Lambda = \lambda_+ + \lambda_- \qquad \textbf{(3.8)}$$

an expression can be written for the conductivity, κ

$$\kappa = \frac{z^2 Fe}{6\pi\eta}\left(\frac{1}{R_+} + \frac{1}{R_-}\right)\frac{\rho}{M_w} \qquad \textbf{(3.9)}$$

where ρ is the density and M_w is the molar mass of the ionic fluid. Therefore, all of the theories developed for limiting molar conductivities in molecular solvents are also applicable to ionic liquids where there is an infinite dilution of suitably sized holes.[27]

It should be noted that equation (3.9) is not applicable to high temperature molten salts because the fraction of suitably sized holes is much larger (c.a. 0.5). The ionic activity therefore becomes an important factor and because there are smaller ions than in the high temperature analogues, the ionic association constant will be significant.

By using equations (3.5) and (3.9), it is possible to predict the minimum viscosity and conductivity that an ionic liquid is able to achieve. In order to obtain an optimum conductivity and viscosity, the surface tension and the size of the ions both need to be as small as possible. This explains why the highest reported conductivity for an ionic liquid to date is that of $[EtNH_3]^+[NO_3]^-$ (c.a. 150 mS cm^{-1} at 298K).[15]

Angell[15] has shown that the molar conductivity is inversely proportional to the viscosity of the liquid for ionic liquids with discrete anions. This correlation is virtually independent of the cation and anion. The properties of ionic liquids have been defined using this method. Since the size of most ionic species that form ionic liquids are approximately the same size ($R \approx$ 3-4 Å), it is clear that equations (3.7) and (3.9) explain Angell's observation, and that a correlation exists between the molar conductivity and the fluidity. Hole theory therefore potentially provides a method by which the dilution of ionic liquids can be quantified because equation (3.9) will no longer be valid when hole mobility does not limit conductivity.

To apply hole theory it is necessary to determine the surface tension of the liquids as this feeds into equation (3.9). The average hole size of the system can be calculated from equation (3.9);

$$4\pi < r^2 >= 3.5\frac{kT}{\gamma} \qquad \textbf{(3.10)}$$

where k is the Boltzmann constant and γ is the surface tension at temperature T. Surface tension could be expected to follow similar trends to viscosity since it is a measure of how strong the intermolecular forces are in the liquid, similar to the measure of viscosity. It is reasonable to expect that as viscosity increases as the molar composition of ChCl increases, the surface tension should increase as the molar

composition of ChCl increases. A plot of surface tension as a function of temperature can be seen in Figure 3.5.

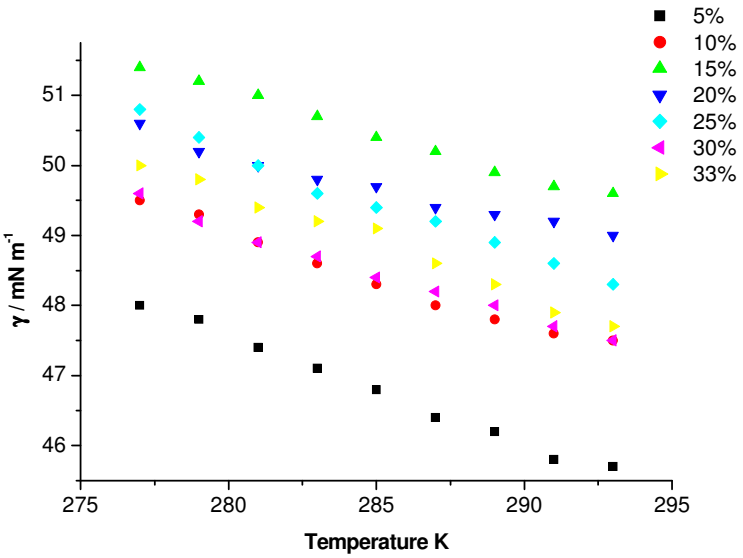


***Figure 3.5**: Plot of temperature versus surface tension as a function of ChCl composition for 1,2-ethanediol systems*

The trend is more obvious in Figure 3.6 where again the data at 293 K alone are plotted.

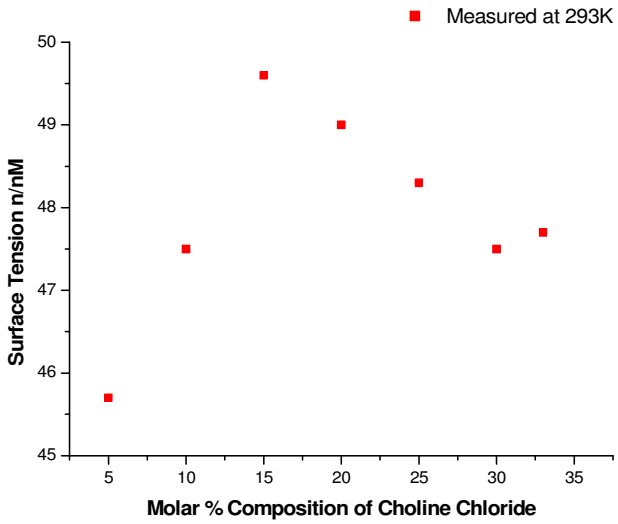


***Figure 3.6**: Plot of surface tension as a function of ChCl composition at 293K for the 1,2-ethanediol systems*

The average hole radius at 293 K is approximately 1.52 Å. The volume of the theoretical mobile complexed species was calculated to be 345.3 Å^3.[28] The assumed complexed species consists of two 1,2-ethanediol molecules (2 x 80.54 Å^3), one choline cation (149.20 Å^3) and one chloride anion (35.02 Å^3). The radius of the mobile species r_s was calculated from the average of the cationic species (i.e. choline) and the anionic species (i.e. chloride and two 1,2-ethanediols). The average radius of mobile species is 3.45 Å. It has previously been shown that there is an empirical correlation between the activation energy for viscous flow and the ratio of the ion an hole size r_i/r_h. Figure 3.7 shows the correlation between E_η and r_i/r_h. From the data above the r_i/r_h for the ChCl: 2 1,2-ethanediol system is 2.27 which would suggest a E_η value of *c.a.* 100 kJ mol^{-1}.

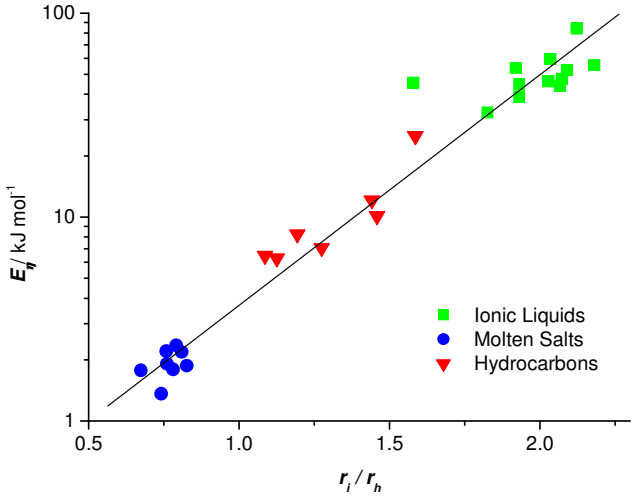


***Figure 3.7**: Energy for activation of viscous flow as a function solvent to hole radius ratio*

Table 3.1 lists the energies of activation of viscosity and conductivity calculated from Figure 3.2 and Figure 3.3. The values for E_η are considerably below 100 kJmol^{-1} and suggest a mobile species which is much smaller that the anion with two bound glycols. Below 25 mol % ChCl the roughly constant E_η value suggests a mobile species of approximately 2.3 Å radius whereas above 30 mol % ChCl the radius is approximately 2.9 Å. Clearly this equates to an average ionic radius but it suggests that there may be a contribution from a free chloride anion (r = 1.81 Å) passing between 1,2-ethanediol molecules, a free 1,2-ethanediol molecule (r = 2.68 Å) or a chloride anion bound to a 1,2-ethanediol molecule (3.03 Å). It seems to suggest that there is a change in mobile species occurring which corresponds to the concentration at which the

conductivity passes through a maximum. FAB-MS shows the presence of the chloride coordinated to both one and two glycol molecules and presumably the composition of the two species will depend upon the composition of the two components in the mixture.

It has previously been shown that for other DESs such as those based on urea or acetamide, the mobile species correspond to the chloride anion bound to two hydrogen bond donors.[7] The results shown above suggest the high conductivities and low viscosities presented in Figures 3.2 and 3.3 arise from smaller mobile species which results from weaker anion-HBD interactions.

***Table 3.1**: Energies of activation of both viscosity and conductivity for 1,2-ethanediol based systems*

Mol. % ChCl	E_η (kJmol^{-1})	ln η_0	E_Λ (kJmol^{-1})	ln κ_0
5	42.28	-14.67	-17.65	8.60
10	39.05	-13.22	-12.82	7.01
15	37.71	-12.56	-14.45	7.91
20	39.26	-13.10	-11.95	6.90
25	35.48	-11.19	-13.37	7.46
30	79.50	-29.15	-12.73	7.11
33	84.48	-31.15	-12.02	6.79

As more of the salt is added to the system it would be expected that the density of the system should increase at a linear rate. This can be seen in Figure 3.8. There is however only a slight change in the density. The addition of 33% molar ratio of ChCl to 1,2-ethanediol results in a small increase of 0.007 gcm^{-3}. This results in an increase of only 0.6% of the density of the pure liquid. This would suggest that there is very little change in the strucutre of the system.

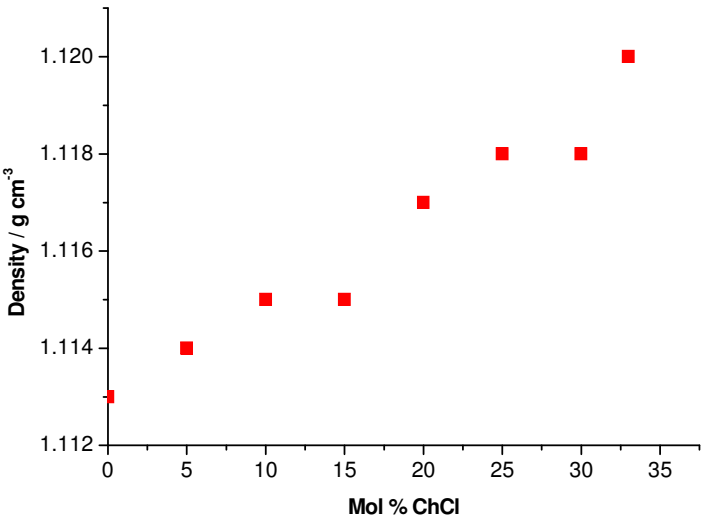


***Figure 3.8**: Plot of molar percentage composition of ChCl versus density at 293K for the 1,2-ethanediol systems*

Using the density data the molar volume V_m can be calculated using

$$V_m = M_r / \rho \quad \textbf{(3.11)}$$

Where M_r is the relative molar mass of the liquid calculated from

$$M_r = x_{ChCl} M_{ChCl} + x_{HBD} M_{HBD} \quad \textbf{(3.12)}$$

where x is the mole fraction and M the relative molar mass of the two components. The actual volume occupied by the components can be calculated in a similar manner i.e.

$$V_{comp} = (x_{ChCl} V_{ChCl} + x_{HBD} V_{HBD})N_A \quad \textbf{(3.13)}$$

Where V is the molecular volume of the component and N_A the Avogadro constant. Using this approach the fractional molar free volume can be obtained using

$$V_{free} = (V_m - V_{comp}) / V_m \quad \textbf{(3.14)}$$

A liner decrease in V_{free} with mole fraction ChCl can be observed in Figure 3.9 representing an almost 3 % contraction of the liquid compared to pure 1,2-ethanediol.

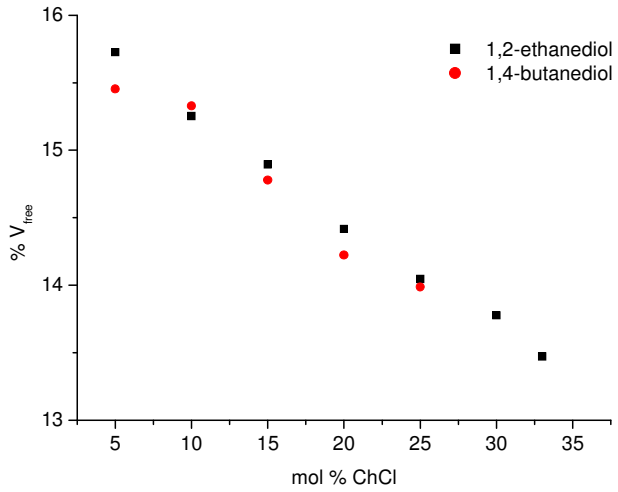


***Figure 3.9**: Plot of V_{free} as a function of mol % ChCl*

3.1.2 1,4-Butanediol and 1,6-Hexanediol

Mixtures of ChCl with 1,4-butanediol (6) were made up ranging from 5% ChCl up to 25% ChCl. It was not possible to go to a higher concentration of ChCl than this as the mixture began to phase separate. The viscosity, conductivity, surface tension and density were all measured and the results are presented below. The results are exactly analogous to those observed for 1,2-ethanediol shown above.

It should be noted that 1,6-hexanediol is a solid at room temperature (melting point 40-43 °C) and so no density data could be obtained due to the high temperature required to form a liquid. The phase diagram of 1,6-hexanediol with ChCl is shown in Figure 3.10.